# Fischer Black, Myron Scholes and Robert C. Merton's Option Pricing Model (BSM)

*A Modern Perspective*

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*Oct 2026*



BLACK SCHOLES MERTON

**The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel**

(often called the Nobel Memorial Prize in Economic Sciences)

*Sveriges Riksbank* is Sweden's central bank and the world's oldest surviving central bank, founded in 1668.

# Questions to address today

| | |
|---|---|
| What were the real contributions of Black-Scholes-Merton? | • NOT the functional form, but the risk-neutral parameters of the model<br>• The continuous-time arbitrage argument (frictionless)<br>• The advancement of the risk-neutrality concept into assets with nonlinear payoffs |
| What are the major drawbacks of the model in practice? | • Forces a lognormal outcome, positively skewed, may or may not fit data<br>• Certainly does not fit SPX data well<br>• The model allows no possibility of zero or negative values of the underlying<br>• The lognormal tails are too thin to accommodate high kurtosis observed in markets |
| What alternatives do we have? | • Specify a proprietary probability distribution (e.g. mixture of normals)<br>• Compute implied probability distribution from traded option prices |

# Earlier attempts at option pricing

*Same general functional form was known*

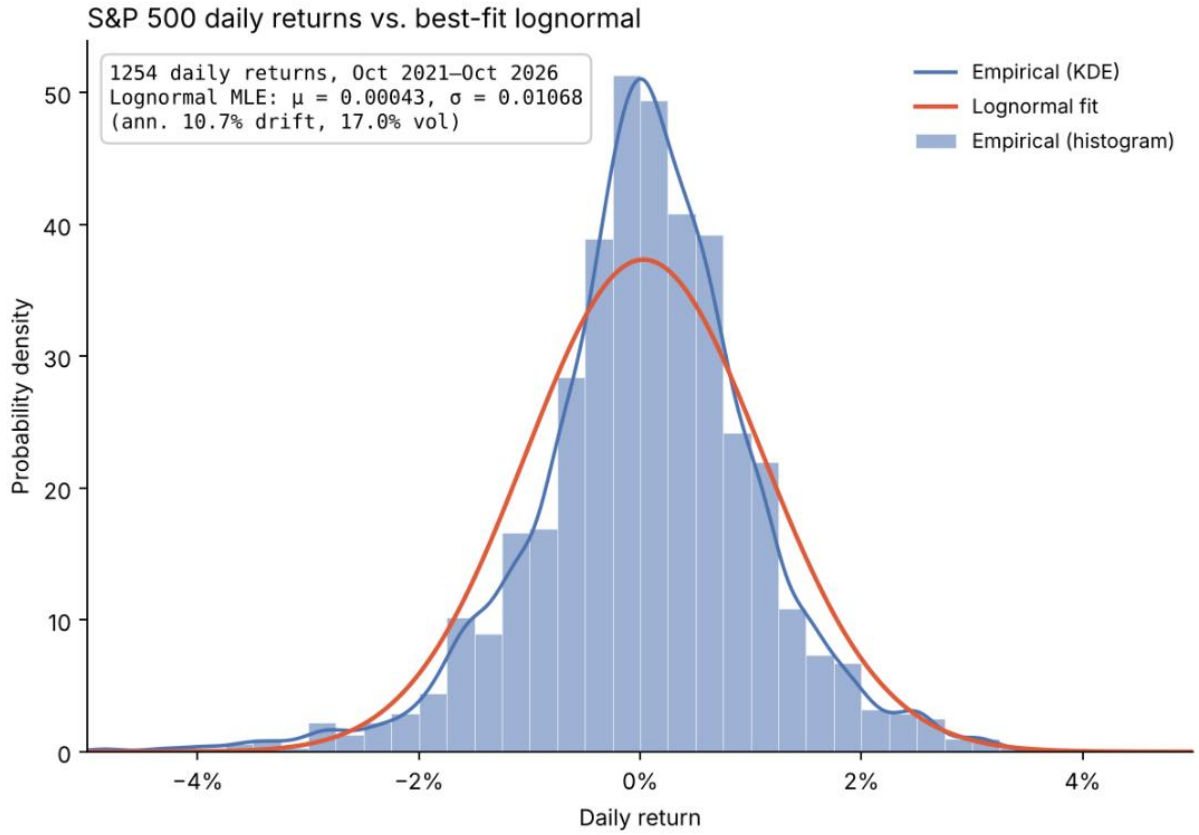
| Author | Price process | Call value | Discounting |
|---|---|---|---|
| Bachelier (1900) | Arithmetic Brownian, zero drift | $C = (S-K)\cdot N(z) + \sigma\sqrt{\tau}\cdot n(z),\ z = (S-K)/\sigma\sqrt{\tau}$ | None; prices can go negative |
| Sprenkle (1961/64) | Lognormal, drift ρ | $C = e^{(\rho\tau)}\cdot S\cdot N(d_1) - (1-Z)\cdot K\cdot N(d_2)$ | None; ad hoc risk factor Z |
| Boness (1964) | Lognormal, drift ρ | $C = S\cdot N(d_1) - e^{(-\rho\tau)}\cdot K\cdot N(d_2)$ | Stock's expected return ρ |
| Samuelson (1965) | Lognormal, drift α | $C = e^{((\alpha-\beta)\tau)}\cdot S\cdot N(d_1) - e^{(-\beta\tau)}\cdot K\cdot N(d_2)$ | Warrant's own rate $\beta \geq \alpha$ |
| Black & Scholes (1973) | Lognormal; drift drops out | $C = S\cdot N(d_1) - e^{(-r\tau)}\cdot K\cdot N(d_2)$ | Risk-free r, by the hedge |

Samuelson nests them all: $\alpha = \beta = \rho$ gives Boness; $\alpha = \beta = r$ gives Black-Scholes. The breakthrough was the hedge, not the integral.

Every published model in 1960-73 assumed a lognormal distribution.

Also, Bachelier assumed the underlying contract was a futures price, already risk-neutral.

# Using the last 5 years of SPX daily returns…



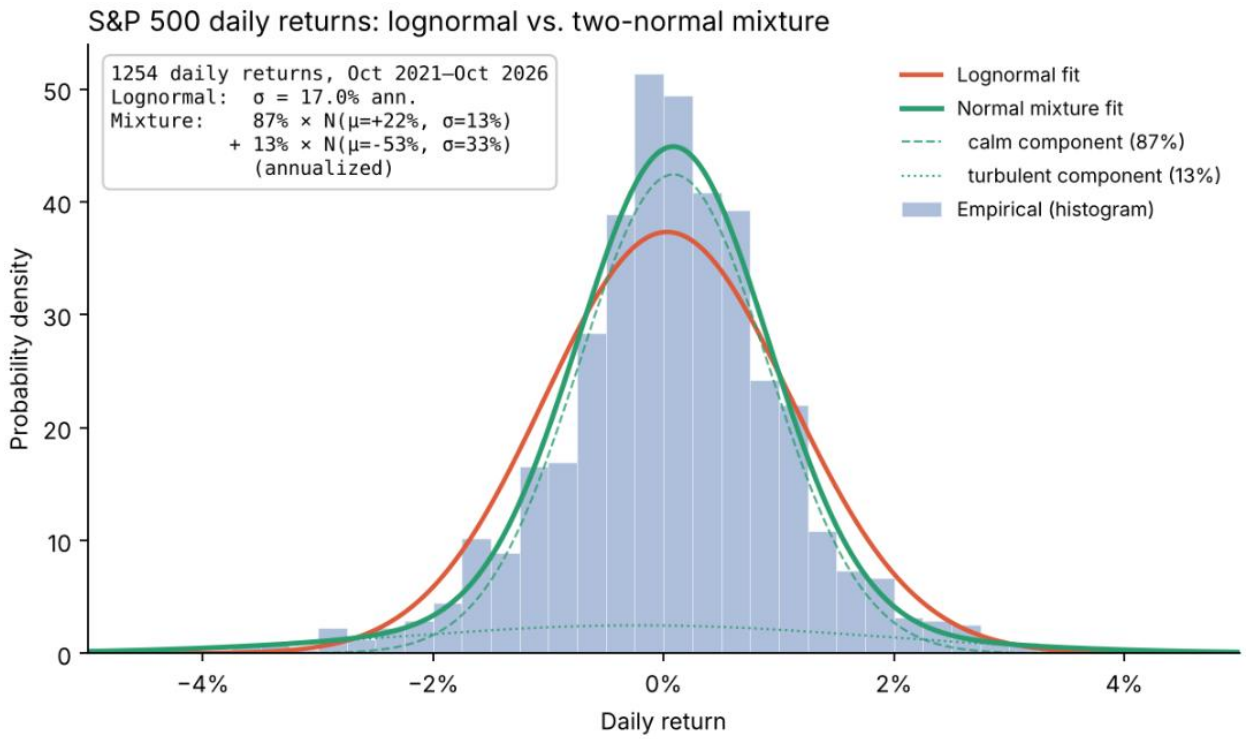

Source: FRED SP500 (S&P 500 index level). Lognormal fit on gross return P_t/P_{t-1}, i.e. ln(P_t/P_{t-1}) ~ N(μ, σ²).

**How well the lognormal fits**

- **Peak and shoulders**
  Empirical peak is much taller and narrower (≈51 vs 37 at the mode), with thinner shoulders around ±1–2%. Excess kurtosis of log returns: 6.4.
- **Tails**
  Moves beyond 3σ / 4σ / 5σ: 13 / 5 / 3 observed vs 3.4 / 0.08 / 0.0007 expected under the lognormal.
- **April 2025 tariff episode**
  Supplies both extremes: −6.0% on Apr 4 and +9.5% on Apr 9.
- **Formal tests**
  Kolmogorov–Smirnov p ≈ 2e-5; Jarque–Bera p ≈ 0. Lognormal decisively rejected.
- **Skew**
  +0.17 on simple returns, ≈ +0.03 on log returns — nearly symmetric, so the miss is almost entirely kurtosis, not skew.

The lognormal distribution does not fit historical returns well.

# A mixture of normals works visibly better



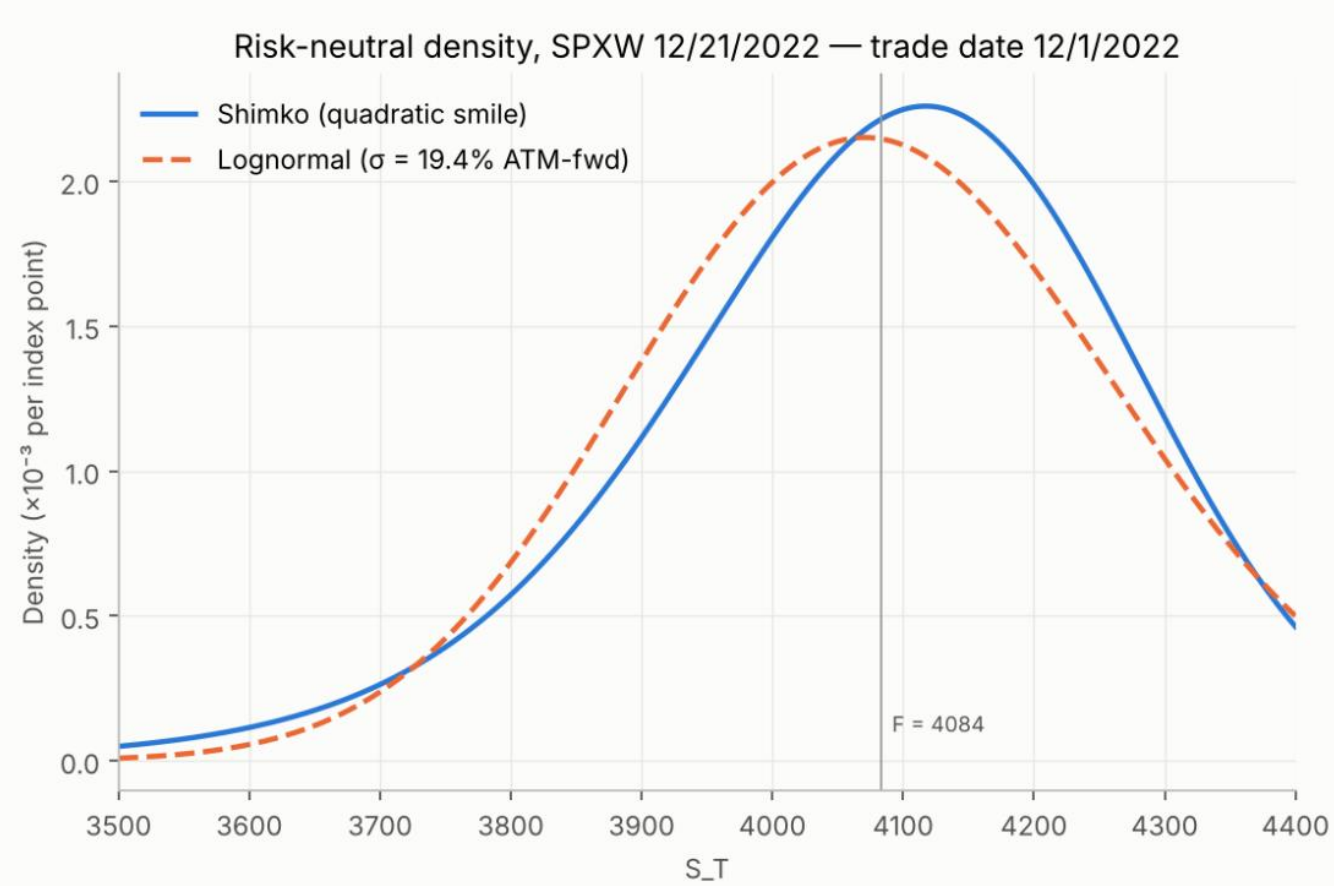

**Lognormal vs. mixture**

- **Overall fit**
  Log-likelihood 3,991 vs 3,912. BIC favors the mixture by 135 points despite 3 extra parameters. Kolmogorov–Smirnov: lognormal rejected ($p \approx 2e\text{-}5$); mixture not rejected ($p = 0.26$).
- **Two regimes**
  87% calm days at ~13% vol with positive drift, 13% turbulent days at ~33% vol with negative drift. That combination produces the tall peak and fat tails.
- **Peak**
  Mixture density at the mode ≈ 45 vs lognormal 37 (KDE ≈ 51). Better, but still a little low.
- **Tails**
  Moves beyond 3σ / 4σ / 5σ: observed 14 / 5 / 3; mixture 20.6 / 6.7 / 1.7; lognormal 3.4 / 0.08 / 0.001. 1-in-1000 down day: empirical −4.7%, mixture −5.2%, lognormal −3.2%.
- **Where it misses**
  Mixture excess kurtosis is 3.5 vs 7.0 in the sample and its skew is −0.30 vs +0.17. Both sample figures are driven by the +9.5% day of Apr 9, 2025, which a likelihood fit largely ignores.

Source: FRED SP500. Lognormal: $\ln(P_t/P_{t-1}) \sim N(\mu, \sigma^2)$ by MLE. Mixture: two-component Gaussian on simple daily returns by EM (MLE). σ multiples use sample s.d. of simple returns.

# How to construct a risk-neutral implied density

(Shimko 1993, “Bounds of Probability”, **RISK** Magazine)




**Shimko vs. lognormal** (3500–4400, no tails)

- Fatter downside tail: P(S<3600) = 1.42% vs 0.30%; P(S<3700) = 3.21% vs 1.60%
- Peak shifted right, above the forward: mode 4117 vs 4071 (F = 4084)
- Less mass between ~3730 and ~4060: P(S<4000) = 30.1% vs 33.3%
- More mass between ~4060 and ~4370
- More negative skew (−0.51 vs −0.22) and higher kurtosis (3.00 vs 2.60)
- Curves meet near 4400, where the quadratic smile flattens

Lognormal σ = 19.44% (ATM-forward vol from the fit)

# Issues with the Shimko 1993 method

## Good

- Provide overall shape of the density
- Used by the Minneapolis FED for years to describe probability distributions of future inflation and interest rates

## Bad

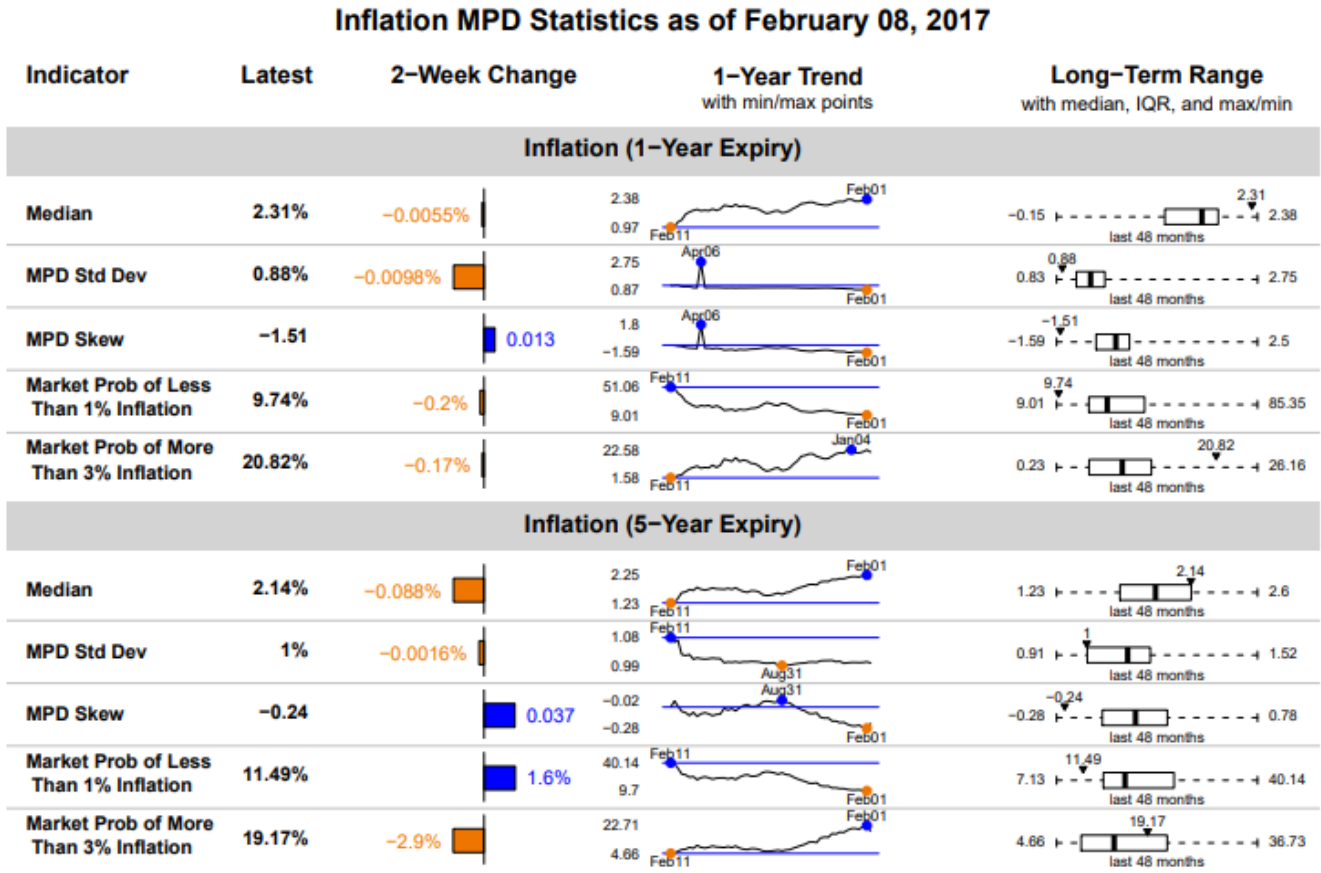
- Cannot extrapolate beyond traded strikes
- The implied volatility curve is used
- Probabilities don’t add to one naturally

## Remedy

- Assume  a distribution or a mixture of distributions

# Sample Minneapolis Federal Reserve Report:

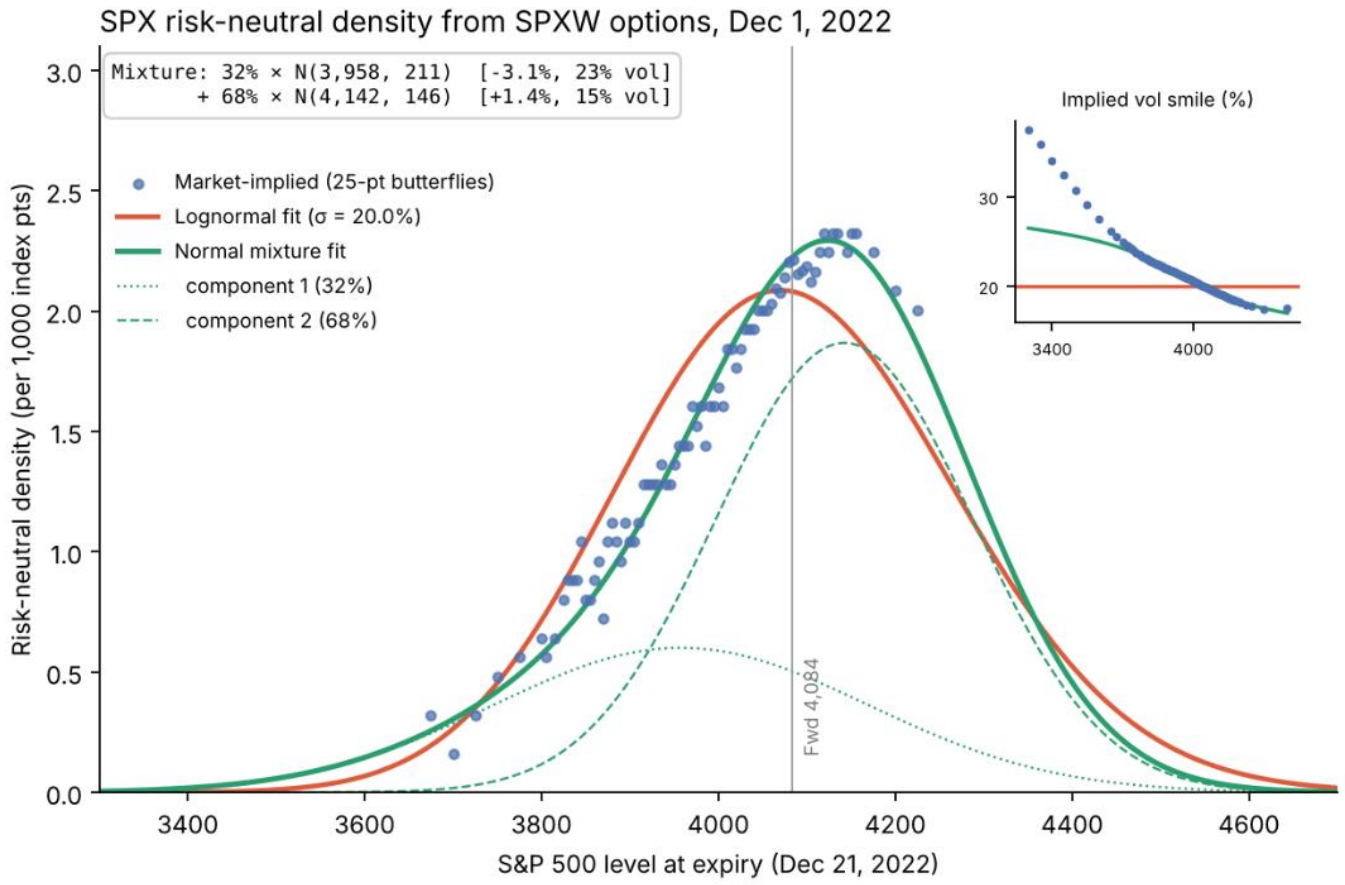
Forecasting future probability distributions of inflation



**Inflation MPD Statistics as of February 08, 2017**

| Indicator | Latest | 2−Week Change | 1−Year Trend (with min/max points) | Long−Term Range (with median, IQR, and max/min) |
|---|---|---|---|---|
| **Inflation (1−Year Expiry)** | | | | |
| Median | 2.31% | −0.0055% | 2.38 / 0.97; Feb01, Feb11 | −0.15 … 2.31 … 2.38; last 48 months |
| MPD Std Dev | 0.88% | −0.0098% | 2.75 / 0.87; Apr06, Feb01 | 0.83 … 0.88 … 2.75; last 48 months |
| MPD Skew | −1.51 | 0.013 | 1.8 / −1.59; Apr06, Feb01 | −1.59 … −1.51 … 2.5; last 48 months |
| Market Prob of Less Than 1% Inflation | 9.74% | −0.2% | 51.06 / 9.01; Feb11, Feb01 | 9.01 … 9.74 … 85.35; last 48 months |
| Market Prob of More Than 3% Inflation | 20.82% | −0.17% | 22.58 / 1.58; Jan04, Feb11 | 0.23 … 20.82 … 26.16; last 48 months |
| **Inflation (5−Year Expiry)** | | | | |
| Median | 2.14% | −0.088% | 2.25 / 1.23; Feb01, Feb11 | 1.23 … 2.14 … 2.6; last 48 months |
| MPD Std Dev | 1% | −0.0016% | 1.08 / 0.99; Feb11, Aug31 | 0.91 … 1 … 1.52; last 48 months |
| MPD Skew | −0.24 | 0.037 | −0.02 / −0.28; Aug31, Feb01 | −0.28 … −0.24 … 0.78; last 48 months |
| Market Prob of Less Than 1% Inflation | 11.49% | 1.6% | 40.14 / 9.7; Feb11, Feb01 | 7.13 … 11.49 … 40.14; last 48 months |
| Market Prob of More Than 3% Inflation | 19.17% | −2.9% | 22.71 / 4.66; Feb01, Feb11 | 4.66 … 19.17 … 36.73; last 48 months |

# Normal Mixture vs. Lognormal for Fitting Traded Market Prices




**Lognormal vs. mixture of normals**

- **Setup**
  110 OTM SPXW mids, 20 days to expiry. Forward 4,083.5 and discount factor 0.99795 (r ≈ 3.7%) from put-call parity. Both models priced with mean fixed at the forward.
- **Fit to prices**
  Lognormal: RMSE $3.43, 3% of options inside bid-ask. Mixture: RMSE $0.37, 43% inside bid-ask; within 3600–4400 the implied-vol error is 0.35 vs 2.3 vol pts.
- **Shape**
  Same total vol (19.9% vs 20.0%), but the mixture puts 32% weight on a −3.1% / 23%-vol state and 68% on a +1.4% / 15%-vol state. Skew −0.49 vs lognormal +0.14.
- **Tail odds**
  Fall >10% by expiry: mixture 2.9%, lognormal 1.3%. Rise >10%: mixture 0.7%, lognormal 2.0%.
- **Where it misses**
  Gaussian tails are too thin beyond about ±10%: deep puts (2600–3500) are worth $0.10–1.60 in the market but ≈$0–0.40 in the mixture, so the smile wings are missed (41% vs 27% IV at 3200).

Source: SPXW quotes 12/1/2022, exp. 12/21/2022. Fits: least squares on OTM mid prices. Market density: second differences of call prices (puts converted via parity), 25-pt spacing.

NOTE: These are risk-neutral densities

SPXW 12/21/2022 calls · trade date 12/1/2022 · F = 4083.5 · r = 3.75% · T = 20 days

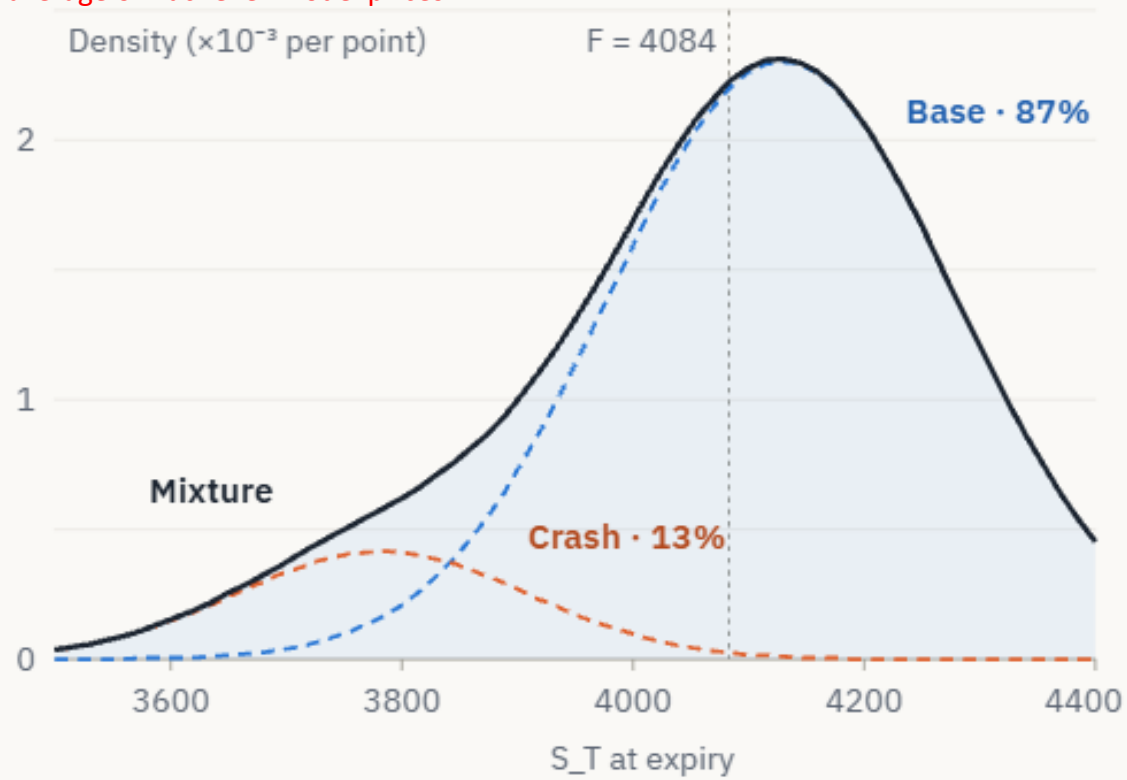
# Fitting a mixture of normals to option prices

NOTE: weighted average of Bachelier model prices

1. **Model** the terminal price: $S_T \sim w \cdot N(\mu_1, \sigma_1^2) + (1-w) \cdot N(\mu_2, \sigma_2^2)$
2. **Price** each call in closed form: $C(K) = e^{-rT} \sum w_i [(\mu_i - K)\Phi(d_i) + \sigma_i \varphi(d_i)]$, with $d_i = (\mu_i - K)/\sigma_i$
3. **Constrain** $\sum w_i \mu_i = F$ so the forward is priced exactly
4. **Fit** $w$, $\mu_i$, $\sigma_i$ by minimizing $\sum [(\hat{C} - C_{mkt})/\text{spread}]^2$ over 98 call mids, K = 3500–4400

| Component | Weight | Mean | SD |
|---|---|---|---|
| Crash | 13.2% | 3783 | 127 |
| Base | 86.8% | 4129 | 151 |

Price RMSE **$0.48**; 97 of 98 strikes fit inside the bid-ask.
Mixture: SD 188, skew −0.41, kurtosis 3.10.




Normal tails are thin below ~3700 (IV misses of 2–8 vol pts); a third component cuts IV RMSE from 120 bp to 29 bp.

# A few questions

Do we need stochastic calculus at all?

Is the lognormal distribution reasonable?

In the BSM model, we have

- Stock Price + unknown stochastic process yields terminal payout distribution.

But we don't really know the stochastic process, and we DO know the option prices

Shouldn't we follow a sequence of knowns rather than unknowns?

- Stock price + Option prices → Implied probability distribution

# Assumptions: Start with Black-Scholes-Merton (1973)

- Parameters

| Parameter | Definition |
|---|---|
| $S_0$ | Current stock price |
| $X$ | Strike price |
| $r$ | Continuously compounded risk-free interest rate |
| $\sigma$ | Annualized stock return volatility |
| $T$ | Time to maturity, in years |
| $\alpha$ | Continuously compounded expected growth rate of the stock price |

We assume that the terminal stock price is

$$S_T = S_0 e^{(\alpha - \sigma^2/2)T + \sigma\sqrt{T}Z}$$

The expected payout of the European call option at time T

$$E[C_T] = S_0 e^{\alpha T} N(d_1) - X N(d_2)$$

$$d_1 = \frac{\ln(S_0/X) + (\alpha + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/X) + (\alpha - \sigma^2/2)T}{\sigma\sqrt{T}}$$

# European Put-Call Parity Seems to Force $\alpha$=r

…This forces $\alpha$=r in part of the formula, but does not affect the $\alpha$ inside the $d_1$ and $d_2$ expressions

$$C = S_0 e^{(\alpha - r)T} N(d_1) - X e^{-rT} N(d_2)$$

$$P = X e^{-rT} N(-d_2) - S_0 e^{(\alpha - r)T} N(-d_1)$$

$$C - P = S_0 e^{(\alpha - r)T} \left[N(d_1) + N(-d_1)\right] - X e^{-rT} \left[N(d_2) + N(-d_2)\right]$$

where, as in (3):

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$$d_1 = \frac{\ln(S_0/X) + (\alpha + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/X) + (\alpha - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

By symmetry of the normal, N(d) + N(−d) = 1, so both brackets equal 1:

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$$C - P = S_0 e^{(\alpha - r)T} - X e^{-rT}$$

Parity holds by static replication with no model, and it requires:

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$$C - P = S_0 - X e^{-rT}$$

Problem fully solved by enforcing monotonicity in the CDF (or convexity in X)

## 1. MODEL

Non-dividend-paying stock, lognormal terminal price with expected growth rate α, so $E[S_T] = S_0 e^{\alpha T}$:

$$S_T = S_0 \exp\left[\left(\alpha - \tfrac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}\,Z\right], \qquad Z \sim N(0,1) \tag{1}$$

## 2. EXPECTED PAYOFF

Integrating the call payoff over the region $S_T > X$:

$$E\big[\max(S_T - X,\, 0)\big] = S_0 e^{\alpha T} N(d_1) - X\,N(d_2) \tag{2}$$

$$d_1 = \frac{\ln(S_0/X) + (\alpha + \tfrac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} \tag{3}$$

## 3. CANDIDATE PRICES

Discount at r. The growth rate appears in two places, so give each its own name: $a$ in the stock leg and $b$ inside $d_1$ and $d_2$. Equation (2) has a = b = α, but we do not assume it.

$$\begin{aligned} C &= S_0 e^{(a-r)T} N(d_1) - X e^{-rT} N(d_2) \\ P &= X e^{-rT} N(-d_2) - S_0 e^{(a-r)T} N(-d_1) \end{aligned} \tag{4}$$

$$d_1 = \frac{\ln(S_0/X) + (b + \tfrac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} \tag{5}$$

## 4. RESTRICTIONS ACROSS STRIKES FORCE A = B

A higher strike cannot be worth more, and a call spread is worth at most the present value of its width:

$$-e^{-rT} \le \frac{\partial C}{\partial X} \le 0 \tag{6}$$

Differentiating (4), and using $S_0 e^{bT}\phi(d_1) = X\phi(d_2)$:

$$\frac{\partial C}{\partial X} = -e^{-rT}\left[N(d_2) - \left(1 - e^{(a-b)T}\right)\frac{\phi(d_2)}{\sigma\sqrt{T}}\right] \tag{7}$$

- **If a < b**, the extra term is positive. Deep out of the money, $N(d_2)$ vanishes faster than $\phi(d_2)$, so $\partial C/\partial X > 0$. Monotonicity fails.
- **If a > b**, the extra term is negative. Deep in the money, $1 - N(d_2)$ vanishes faster than $\phi(d_2)$, so $\partial C/\partial X < -e^{-rT}$. The call-spread bound fails.

$$a = b \tag{8}$$

Only then is $-e^{rT}\,\partial C/\partial X = N(d_2)$ a probability, so that C is a genuine discounted expectation (Breeden–Litzenberger).

## 5. RESTRICTIONS TIED TO THE STOCK FORCE A = R

**Zero-strike call.** A call with X = 0 pays $S_T$, so it is the stock. As $X \to 0$, $N(d_1), N(d_2) \to 1$ for any b:

$$\lim_{X\to 0} C = S_0\, e^{(a-r)T} = S_0 \tag{9}$$

**Price bounds.** The call must satisfy

$$\max\left(S_0 - X e^{-rT},\, 0\right) \le C \le S_0 \tag{10}$$

Near X = 0, C equals $S_0 e^{(a-r)T}$. If a > r it exceeds $S_0$ (sell the call, buy the stock). If a < r it falls below $S_0 - X e^{-rT}$ (buy the call, short the stock, lend).

**Put-call parity.** From (4):

$$\begin{aligned} C &= S_0 e^{(a-r)T} N(d_1) - X e^{-rT} N(d_2) \\ P &= X e^{-rT} N(-d_2) - S_0 e^{(a-r)T} N(-d_1) \\ C - P &= S_0 e^{(a-r)T}\left[N(d_1) + N(-d_1)\right] - X e^{-rT}\left[N(d_2) + N(-d_2)\right] \\ &= S_0 e^{(a-r)T} - X e^{-rT} \end{aligned} \tag{11}$$

Static replication requires

$$C - P = S_0 - X e^{-rT} \tag{12}$$

Each of (9), (10) and (12) holds only if

$$a = r \tag{13}$$

None of them constrains b: the terms in N cancel or go to 1 whatever drift sits inside $d_1$ and $d_2$.

## 6. CONCLUSION

Combining (8) and (13) gives $b = a = r$, so α = r in both places. Substituting into (4) and (5):

$$C = S_0 N(d_1) - X e^{-rT} N(d_2) \tag{14}$$

$$d_1 = \frac{\ln(S_0/X) + (r + \tfrac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} \tag{15}$$

**Neither family of restrictions is enough alone. Restrictions across strikes make the formula a discounted expectation; restrictions tied to the stock set that expectation's drift to r.**

# Bachelier: The model works for normally distributed underlying

### 1. MODEL

Normally distributed terminal price with mean $S_0 e^{aT}$ and standard deviation $\sigma\sqrt{T}$, where σ is in price units and $S_0 > 0$:

$$S_T = S_0 e^{aT} + \sigma\sqrt{T}\,Z, \qquad Z \sim N(0,1) \tag{1}$$

### 2. EXPECTED PAYOFF

The call pays off when $Z > -d$. Integrating, and using $\int_{-d}^{\infty} z\,\phi(z)\,dz = \phi(d)$:

$$E\big[\max(S_T - X,\,0)\big] = \big(S_0 e^{aT} - X\big)N(d) + \sigma\sqrt{T}\,\phi(d) \tag{2}$$

$$d = \frac{S_0 e^{aT} - X}{\sigma\sqrt{T}} \tag{3}$$

### 3. CANDIDATE PRICES

Discount at r. The growth rate appears in two places, so give each its own name: $a$ in the stock leg and $b$ inside d. Equation (2) has a = b = α, but we do not assume it.

$$\begin{aligned} C &= S_0 e^{(a-r)T} N(d) - X e^{-rT} N(d) + \sigma\sqrt{T}\,e^{-rT}\phi(d) \\ P &= X e^{-rT} N(-d) - S_0 e^{(a-r)T} N(-d) + \sigma\sqrt{T}\,e^{-rT}\phi(d) \end{aligned} \tag{4}$$

$$d = \frac{S_0 e^{bT} - X}{\sigma\sqrt{T}} \tag{5}$$

### 4. RESTRICTIONS ACROSS STRIKES FORCE A = B

A higher strike cannot be worth more, and a call spread is worth at most the present value of its width:

$$-e^{-rT} \;\le\; \frac{\partial C}{\partial X} \;\le\; 0 \tag{6}$$

Differentiating (4), and using $\phi'(d) = -d\,\phi(d)$ and $d\,\sigma\sqrt{T} = S_0 e^{bT} - X$:

$$\frac{\partial C}{\partial X} = -e^{-rT}\left[N(d) - S_0\big(e^{bT} - e^{aT}\big)\frac{\phi(d)}{\sigma\sqrt{T}}\right] \tag{7}$$

- **If a < b**, the extra term is positive. Deep out of the money, $N(d)$ vanishes faster than $\phi(d)$, so $\partial C/\partial X > 0$. Monotonicity fails.
- **If a > b**, the extra term is negative. Deep in the money, $1 - N(d)$ vanishes faster than $\phi(d)$, so $\partial C/\partial X < -e^{-rT}$. The call-spread bound fails.

$$a = b \tag{8}$$

Only then is $-e^{rT}\,\partial C/\partial X = N(d)$ a probability, and $e^{rT}\,\partial^2 C/\partial X^2 = \phi(d)/(\sigma\sqrt{T})$ the normal density of $S_T$ (Breeden–Litzenberger).

### 5. RESTRICTIONS TIED TO THE UNDERLYING FORCE A = R

**Deep in-the-money call.** Since $S_T$ can be negative, a zero-strike call is no longer the underlying. Instead, as $X \to -\infty$ the call is exercised with certainty and becomes a forward contract, worth $S_0 - Xe^{-rT}$. In (4), $N(d) \to 1$ and $\phi(d) \to 0$ for any b:

$$C \;\to\; S_0 e^{(a-r)T} - X e^{-rT} = S_0 - X e^{-rT} \tag{9}$$

**Price bounds.** The upper bound $C \le S_0$ fails here, because the call pays 0 when $S_T < 0$. The two lower bounds still hold, since each payoff is at least the corresponding forward payoff:

$$\begin{aligned} C &\ge S_0 - X e^{-rT} \\ P &\ge X e^{-rT} - S_0 \end{aligned} \tag{10}$$

If a < r, the call approaches $S_0 e^{(a-r)T} - Xe^{-rT}$ as $X \to -\infty$, which is below its bound (buy the call, short the underlying, lend). If a > r, the put approaches $Xe^{-rT} - S_0 e^{(a-r)T}$ as $X \to +\infty$, which is below its bound (buy the put, buy the underlying, borrow).

**Put-call parity.** From (4):

$$\begin{aligned} C &= S_0 e^{(a-r)T} N(d) - X e^{-rT} N(d) + \sigma\sqrt{T}\,e^{-rT}\phi(d) \\ P &= X e^{-rT} N(-d) - S_0 e^{(a-r)T} N(-d) + \sigma\sqrt{T}\,e^{-rT}\phi(d) \\ C - P &= \big(S_0 e^{(a-r)T} - X e^{-rT}\big)\big[N(d) + N(-d)\big] \\ &= S_0 e^{(a-r)T} - X e^{-rT} \end{aligned} \tag{11}$$

Static replication requires

$$C - P = S_0 - X e^{-rT} \tag{12}$$

Each of (9), (10) and (12) holds only if

$$a = r \tag{13}$$

None of them constrains b: the volatility terms cancel or vanish, and the N terms sum or go to 1 whatever drift sits inside d.

### 6. CONCLUSION

Combining (8) and (13) gives $b = a = r$, so α = r in both places. Substituting into (4) and (5):

$$C = e^{-rT}\Big[\big(S_0 e^{rT} - X\big)N(d) + \sigma\sqrt{T}\,\phi(d)\Big] \tag{14}$$

$$d = \frac{S_0 e^{rT} - X}{\sigma\sqrt{T}} \tag{15}$$

**This is the Bachelier price on the forward $F = S_0 e^{rT}$. The two families of restrictions play the same roles as in the lognormal case: restrictions across strikes make the formula a discounted expectation, and restrictions tied to the underlying set that expectation's mean to the forward.**

# European option pricing for any terminal probability distribution

## 1. SETUP

The bond, the underlying, and European calls and puts at all strikes trade for one maturity T. A candidate pricing rule discounts at rate k under a distribution Q with density q:

$$C(X) = e^{-kT} E^Q[(S_T - X)^+]$$
$$P(X) = e^{-kT} E^Q[(X - S_T)^+] \quad (1)$$

## 2. RESTRICTIONS ACROSS STRIKES FORCE A SINGLE Q

Differentiating under the expectation:

$$\frac{\partial C}{\partial X} = -e^{-kT} Q(S_T > X), \qquad \frac{\partial^2 C}{\partial X^2} = e^{-kT} q(X) \quad (2)$$

When Q is a genuine probability, C is decreasing and convex with slope between $-e^{-kT}$ and 0. Conversely (Breeden–Litzenberger, Carr–Madan), any call price function with these properties determines exactly one Q:

$$Q(S_T > X) = -e^{kT} \frac{\partial C}{\partial X}, \qquad q(X) = e^{kT} \frac{\partial^2 C}{\partial X^2} \quad (3)$$

This generalizes a = b. If the stock leg $E[S_T;\ S_T > X]$ and the probability leg $P(S_T > X)$ come from different distributions, (3) does not return a survival function and a density. Somewhere the slope leaves its bounds or the curvature turns negative.

## 3. THE BOND FORCES K = R

A call spread from the lowest to the highest strike pays 1 in every state, so it is the zero-coupon bond. By (2):

$$C(X_{\text{low}}) - C(X_{\text{high}}) \to e^{-kT} = e^{-rT} \implies k = r \quad (4)$$

In the lognormal and Bachelier cases, discounting at r was a hypothesis. In general it is forced once the bond trades.

## 4. RESTRICTIONS TIED TO THE UNDERLYING FIX THE MEAN

For any Q, put-call parity in (1) is an identity:

$$C - P = e^{-rT}(E^Q[S_T] - X) \quad (5)$$

Static replication requires

$$C - P = S_0 - Xe^{-rT} \quad (6)$$

so

$$E^Q[S_T] = S_0 e^{rT} \quad (7)$$

With support on $[0, \infty)$, the zero-strike call gives the same condition: $C(0) = e^{-rT} E^Q[S_T] = S_0$. With support on the whole line, the deep in-the-money limit $X \to -\infty$ gives it.

## 5. THEOREM

The prices of the bond, the underlying, and all calls and puts are free of static arbitrage if and only if there is a probability distribution Q such that

$$C(X) = e^{-rT} E^Q[(S_T - X)^+] \quad \text{and} \quad E^Q[S_T] = S_0 e^{rT} \quad (8)$$

This is the static, single-maturity form of the fundamental theorem of asset pricing. Static arbitrage pins down exactly two moments of Q:

$$\int q(x)\,dx = 1 \ \text{(priced by the bond)}, \qquad \int x\,q(x)\,dx = S_0 e^{rT} \ \text{(priced by the underlying}$$

## 6. Q MUST BE EQUIVALENT TO THE REAL-WORLD DISTRIBUTION

Let $\mathbb{P}$ be the real-world distribution of $S_T$. A butterfly spread concentrated on an interval A pays a positive amount on A and nothing elsewhere.

- If $\mathbb{P}(A) > 0$ but $Q(A) = 0$, the butterfly costs nothing and pays off with positive probability. Buy it.
- If $Q(A) > 0$ but $\mathbb{P}(A) = 0$, the butterfly has a positive price and never pays. Sell it.

$$Q(A) = 0 \iff \mathbb{P}(A) = 0 \quad (10)$$

## 7. WHAT THIS SAYS ABOUT A

- **"α = r" becomes "the Q-mean equals the forward."** The real-world growth rate α never enters any price.
- **Nothing else about Q is determined statically.** Variance, skew and tails of Q are free, subject only to (10).
- **The lognormal and Bachelier results add a modeling assumption.** Each assumed Q is $\mathbb{P}$ with only the mean shifted, so the whole adjustment was α → r. Dynamic replication in Black-Scholes justifies this, since the change of measure shifts the drift and leaves σ unchanged. In general the change from $\mathbb{P}$ to Q can reshape the whole distribution.

| MODEL | Q-MEAN OF $S_T$ | CONDITION (7) GIVES |
|---|---|---|
| Lognormal | $S_0 e^{\alpha T}$ | α = r |
| Bachelier | $S_0 e^{\alpha T}$ | α = r |
| General Q | $E^Q[S_T]$ | $E^Q[S_T] = S_0 e^{rT}$ |

**Static arbitrage fixes the total mass of Q through the bond and its mean through the forward. Everything else must come from market option prices or a dynamic hedging argument.**